

MTH212M ASSIGNMENT 2

Instructions:

- (i) Each question is of 10 marks.
- (ii) Only one question will be evaluated. The question number (to be evaluated) will be declared after the deadline for submission.
- (iii) Deadline for submission: **10:00 am IST, Friday February 13, 2026.**
- (iv) **Assignment submission will be through the HelloIITK portal.** Email submissions and any late submission through the HelloIITK portal will NOT be graded.
- (v) Please **upload one single PDF file** containing all your answers.
- (vi) Provide all details/justifications.

Question 1. (a) (1 + 1 + 1 marks) Prove or give a counter-example for each of the following statements.

- (i) Suppose that $i \rightarrow j$. If i is transient, then j is transient.
 - (ii) Suppose that $i \rightarrow j$. If i is aperiodic and positive recurrent, then j is aperiodic and positive recurrent.
 - (iii) Suppose that $i \rightarrow j$. If i is aperiodic and null recurrent, then j is aperiodic and null recurrent.
- (b) (1 + 1 + 1 + 1 + 3 marks) Let $M > 2$ be a positive integer. Let $\{X_n\}_{n=0}^\infty$ be a Markov Chain on the state space $S = \{1, 2, 3, \dots, M\}$ and with the T.P. matrix $P = (p_{ij})_{i,j \in S}$ such that $p_{ij} > 0, \forall 1 \leq i \leq j \leq M$ and 0 otherwise. Which of the following statements are necessarily true?
- (i) There is a single communication class.
 - (ii) There is an absorbing state.
 - (iii) All states have period 2.
 - (iv) P^T , the transpose of P , is stochastic.
 - (v) The sequence $\{X_n^2\}_{n=0}^\infty$ forms a Markov Chain.

Please turn over

Question 2. ((2 + 2) + 6 marks) Let $\{X_n\}_{n=0}^{\infty}$ be a Markov Chain on the state space $S = \{0, 1, 2, 3, 4, 5\}$ and with the T.P. matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{3}{4} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

- (a) Find all transient states and recurrent classes.
- (b) For all transient states i and all recurrent classes C , compute

$$\mathbb{P}(X_n \in C \text{ for some } n \mid X_0 = i).$$